

EEG-NIRS dataset TYUT emotion recognition (ENTER)

Table1 Information about the video stimuli

Clip Number	Source Film	Targeted Emotion	Duration (s)	Language
1	Aftershock	Sad	87	Chinese
2	Tangshan Earthquake	Sad	95	Chinese
3	The Bravest	Sad	73	Chinese
4		Sad	59	Chinese
5	ying to Survive	Sad	100	Chinese
6		Sad	65	Chinese
7	A Little Red Flower	Sad	47	Chinese
8	The Allure of Tears	Sad	90	Chinese
9	Run For Young	Sad	81	Chinese
10	Heaven calls	Sad	83	Chinese
11	A Little Reunion	Sad	82	Chinese
12		Sad	87	Chinese
13	With You	Sad	43	Chinese
14		Sad	116	Chinese
15		Sad	78	Chinese
16	Trump Card (season V)	Happy	113	Chinese
17		Happy	116	Chinese
18		Happy	82	Chinese
19		Happy	118	Chinese
20		Happy	57	Chinese
21		Happy	63	Chinese
22		Happy	117	Chinese
23		Happy	81	Chinese
24		Happy	54	Chinese
25		Happy	42	Chinese
26		Happy	83	Chinese
27		Happy	42	Chinese
28		Happy	56	Chinese
29		Happy	102	Chinese
30		Happy	60	Chinese
31	The Tale of Chinese Medicine (season I)	Calm	58	Chinese
32		Calm	90	Chinese
33		Calm	84	Chinese
34		Calm	92	Chinese
35		Calm	100	Chinese
36		Calm	60	Chinese
37		Calm	63	Chinese
38		Calm	95	Chinese
39		Calm	68	Chinese

40		Calm	70	Chinese
41		Calm	71	Chinese
42		Calm	90	Chinese
43		Calm	87	Chinese
44		Calm	86	Chinese
45		Calm	77	Chinese
46	Bedfellows	Fear	83	English
47	Annabelle	Fear	56	English
48		Fear	67	English
49	Lights Out	Fear	85	English
50	A Quiet Place	Fear	62	English
51	It	Fear	112	English
52	Pictured	Fear	61	English
53	Under the Bed	Fear	51	English
54	The Shining	Fear	75	English
55	Closet Space	Fear	76	English
56	Unfriend	Fear	73	English
57	Creeping	Fear	76	English
58	Dead Air	Fear	70	English
59	The Mime	Fear	74	English
60	The Mime	Fear	81	English

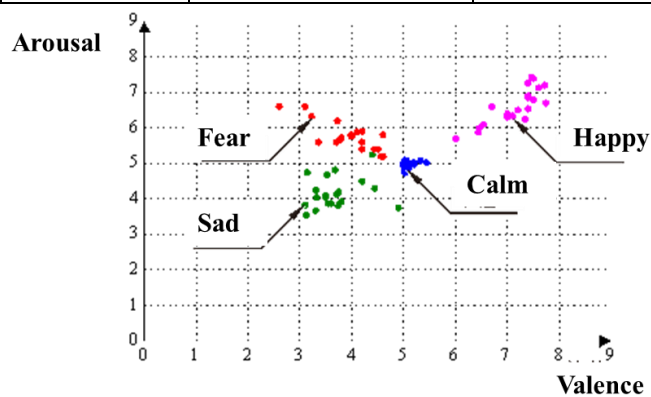


Figure 1. Stimulus material pre-assessment



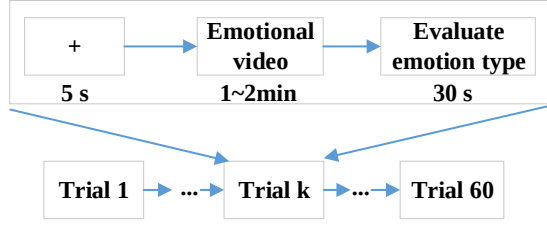


Figure 2. Experimental paradigm.

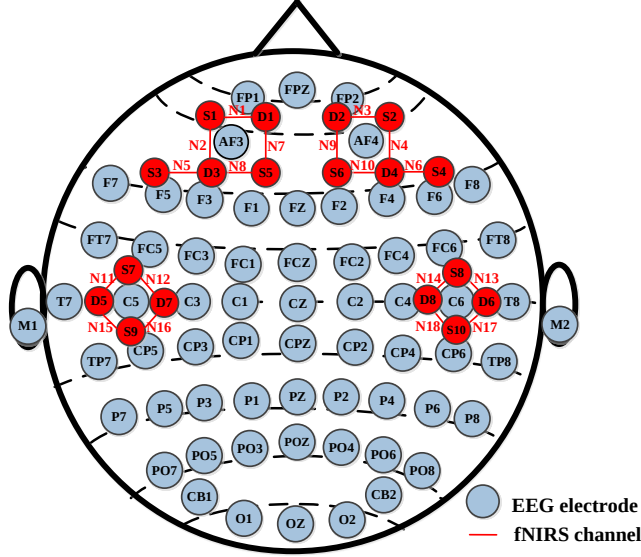


Figure 3. EEG and fNIRS channel distribution map.

The EEG-NIRS dataset TYUT emotion recognition (ENTER), is a collection of EEG-fNIRS datasets evoked by four kinds of emotion video clips. Fifty college students including 25 male (age: 22.92 ± 1.71) and 25 female (age: 24.12 ± 1.67) volunteered for this study. All participants self-reported normal or correct-ed-to-normal vision and normal hearing in the experiments. Each participant gave written informed consent prior to participation, and all of them self-identified as right-handed and self-reported to have no history of mental illnesses or drug, which is the inclusion criteria. The EEG-fNIRS data from all participants were included in the data analysis. Each participant was asked to sit on a comfortable chair facing a computer screen and assigned to watch 60 emotional video clips including sad, happy, calm and fear, and each kind of emotion has 15 video clips. The information about the video stimuli is listed in Table 1. The selected video clips were pre-assessed from the dimensions of arousal and valence according to scores (1-9) from 20 sub-jects, who are not the experimental subjects, as shown in Figure 1. The experimental protocol is shown in Figure 2. Each video clip lasted 1~2 minutes, and then the participant can evaluate the type of emotion within 30s by filling out an evaluation form. Each participant was instructed in the experimental procedure in detail before performing the experiment.

Both EEG and fNIRS sensors were placed on the participant's scalp, and there was no contact between these two hardware devices (Neuroscan SynAmps2 and DanYang HuiChuang NirSmart). The EEG was recorded with 1000 Hz sampling rate and 64 channels, and the fNIRS with 11Hz sampling rate and 18 channels. EEG and fNIRS channel distribution map is shown in Figure 3. During the experiment, each participant was required to minimize his head movements to avoid

signal artifacts.

ENTER is open for the academic community. If you are interested in the dataset and want to gain the access to it, please download and fill out the license agreement below and send it to chenguijun@tyut.edu.cn. We will send you a download link through email after a review of your application.